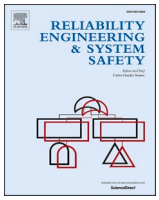
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Probabilistic framework for seismic resilience assessment of transmission tower-line systems subjected to mainshock-aftershock sequences

Juncai Liu, Li Tian *, Meng Yang, Xiangrui Meng

Shandong Research Institute of Industrial Technology, Jinan, Shandong 250098, China

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ABSTRACT

Immediate damage to electricity transmission systems after strong earthquakes can significantly cause widespread power outages and impede maintenance activities. Whereas previous seismic resilience assessments have investigated comprehensively the recovery capability of structures under isolated mainshocks, the effect of aftershocks has received less attention. This paper introduces a probabilistic framework for resilience assessment incorporating the effect of aftershocks and utilizes it to investigate the seismic performance of a transmission tower-line system (TTLS) under mainshock-aftershock (MSAS) sequences. Exceedance probabilities for different damage states of the TTLS are analyzed using fragility curves. Multiple decision variables (including functionality loss, recovery time, recovery function and resilience indicator) are subsequently calculated to quantify the effect of aftershocks. Meanwhile, the seismic resilience of the TTLS is further assessed by considering the uncertainties associated with those decision variables. The results reveal that aftershocks exert a significant influence on the seismic resilience of the TTLS, particularly when it experiences minor or moderate damage following the mainshocks. However, as the damage to the TTLS worsens due to larger mainshocks, the role of subsequent aftershocks becomes comparatively less influential. The proposed framework provides valuable insights for infrastructure systems before and after earthquake, effectively mitigating the adverse effects of sequential earthquakes.

1. Introduction

Electricity transmission systems play a crucial role in modern society as they undertake the transporting and distributing activities of electric energy from power plants to end-users. These systems are widely recognized as essential components of critical infrastructures, contributing significantly to the high-quality development of economies and societies. However, transmission tower-line systems (TTLS) are particularly vulnerable to extreme events especially strong earthquakes during the service life, due to their characteristics of high flexibility, large span, and strong nonlinearity. Any resultant damage or failure of TTLSs can directly cause large-scale interruption of the power grids, leading to enormous economic losses and impeding post-seismic recovery efforts. For instance, the 2008 Wenchuan earthquake resulted in approximate 2700 TTLSs losing their functionality, leading to over 2.46 million costumers experiencing power outages and economic losses exceeding 40 billion RMB [1,2]. More severe damage to the electricity transmission systems was observed during Great East Japan Earthquake in March 2011, leaving 8.5 million customers without power. The restoration works of the entire power grids took over 30 months [3]. Another example occurred in November 2016 when a magnitude 7.8 earthquake struck the upper South Island of New Zealand, tripping a number of TTLSs in the surrounding areas and causing around 7000 businesses and homes without electricity [4]. Therefore, A rapid recovery of damaged TTLSs after earthquakes is of critical importance. It not only improves power transmission efficiency but also facilitates the recovery of other essential electricity-dependent infrastructures such as communication, medical, and energy systems. To mitigate the seismic-induced damage to TTLSs and minimize the negative impacts on both social and economic dimensions, it is crucial to comprehensively estimate the seismic performances of TTLSs.

Seismic resilience is an important metric for assessing the effectiveness of disaster management, and it has received numerous attentions within the field of civil engineering [5–9]. There is a broadly accepted concept that defines the seismic resilience as the ability of an engineering structure to withstand strong earthquakes, minimize socio-economic disruption, and quickly recover normal functionality. Seismic resilience mainly has four attributes: robustness, rapidity,

* Corresponding author.
E-mail address: tianli@sdu.edu.cn (L. Tian).

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redundancy, and resourcefulness [10]. Although its conceptual clarity, quantifying the assessment process of seismic resilience remains complex. Therefore, a reasonably quantifiable evaluation of the seismic resilience of structures or systems can provide comprehensive insight into the impact of earthquakes and assist decision-makers in formulating optimal management strategies [11,12]. The assessment of the seismic resilience establishes a vital link between the structural failure and its economic consequences, while also describing critical decision variables such as functionality losses and recovery strategies. Considering that the damage or failure of each structural component causes countable economic losses, using seismic economic loss to represent structural functionality loss is a reasonable approach [13]. Seismic economic loss mainly consists of direct and indirect losses. Direct loss refers to the costs associated with repairing or rebuilding a damaged structure and can be represented as a dimensionless factor calculated by weighting loss ratios at different damage states. On the other hand, indirect loss is caused by a reduction in the service level of post-earthquake structures, and gradually decreases as repair work progresses. Reduced economic loss reflects the excellent robustness and redundancy of a structure in responding to seismic impacts [14]. Functionality recovery pertains to the process of recovering the functionality of earthquake-damaged structures through maintenance or reinforcing measures [15]. The post-recovery functionality may be less than, equal to, or greater than the pre-earthquake functionality, depending on the retrofitting strategies and the extent of structural damage states [16]. Describing the recovery process is challenging due to the need for a comprehensive consideration of uncertainties in social and economic dimensions. Current resilience assessment frameworks generally utilize time-functionality analysis models to simplify and quantify the recovery process of post-earthquake structures. Several analysis models have been developed [17], including linear, trigonometric, and exponential models. The exponential model is adopted when the social response is propelled by the prompt allocation of adequate resources, while the trigonometric model is used when there is a deficiency of preparedness and sources [18]. In view of the remarkable benefits of incorporating the resilience assessment framework in various systems such as healthcare [19,20], power [21], traffic [22,23], and water supply [24] systems, it is essential to integrate it into the seismic resilience assessment of TTLSs.

Existing seismic resilience assessment frameworks primarily focus on major seismic events with long reference periods, known as mainshocks, which are often followed by a series of aftershocks. Aftershocks, although typically smaller magnitudes, may exhibit longer durations, larger peak ground accelerations, and different spectral characteristics [25]. A recent noteworthy example is the Türkiye earthquake in February 2023, in which two devastating earthquakes with magnitudes of 7.8 and 7.6 occurred on the same day, resulting in significant human casualties and enormous economic losses. Over the next 15 days, more than 5000 aftershocks occurred, which caused great difficulties in restoring the post-earthquake functionalities of infrastructures [26]. In most seismic events, the delay between the mainshock and the largest aftershock may be range between several days to months, thus it is impractical to fully repair a mainshock-damaged structure to its pre-earthquake state before the occurrence of subsequent aftershocks [27,28]. In light of this, Li et al. [29] and Wen et al. [30] investigated the effect of aftershocks on the collapse and cumulative damage of structures, discovering a higher probability of failure for the mainshock-damaged structures in the presence of aftershocks. Furthermore, many investigators [31–35] have assessed the seismic resilience of various structures considering the effect of aftershocks. Their findings indicated that aftershocks can exacerbate economic losses and prolong the disruption of functionality. However, these studies have only assumed that there is no improvement in the functionality during the time gap between the mainshock and aftershock, without taking into account the effect of incomplete functional recovery prior to the aftershocks on the seismic performances of structures.

As for the TTLS, it is of utmost importance that it should possess the ability to withstand not only a single mainshock but also the impact of multiple aftershocks. Previous studies by Liu et al. [36] and Kassem et al. [37] have investigated the probabilistic seismic performance of TTLSs under sequential earthquakes and found that aftershocks can significantly contribute to accumulative damage to TTLSs. Their research highlights the importance of accounting for the effect of aftershocks when assessing the seismic performance of TTLSs. However, to the best of the author's knowledge, there is no available framework yet to incorporate aftershocks into the seismic evaluation of TTLSs, particularly from a resilience perspective.

Based on the above discussions, the main objective of this paper is to evaluate the seismic resilience of a TTLS considering the effect of aftershocks. For this, this paper carries out the following investigations: in Section 2, a seismic resilience assessment framework that incorporates the effect of aftershocks is proposed; a finite element (FE) model of a TTLS is developed using ABAQUS software in Section 3, and a group of as-recorded MSAS sequences are selected from an international ground motion database; in Sections 4 and 5, the seismic resilience of the TTLS is thoroughly investigated through an in-depth analysis of functionality loss, recovery time, recovery function, and resilience indicators; and the main conclusions are synthesized in Section 6.

2. Seismic resilience assessment framework

The seismic resilience assessment framework consists of three blocks: nonlinear dynamic analysis, structural fragility analysis, and seismic resilience assessment, as illustrated in Fig. 1. In the first block, an FE model of a TTLS is constructed using ABAQUS software and 15 pairs of as-recorded MSAS sequences are selected from PEER Strong Motion Database [38]. Subsequently, nonlinear dynamic analyses of the TTLS are performed to obtain the probabilistic seismic demand model, which describes the distribution of the engineering demand parameters (*EDP*) of the TTLS at a given earthquake intensity measure (*IM*). Assuming a log-normal distribution for those analytical data, the exceedance probabilities for different damage states (DS_i) of the TTLS are determined under a given *IM* and typically visualized through fragility curves. Finally, through the consideration of multiple decision variables including functionality loss, recovery time, recovery function, and resilience indicator, a comprehensive assessment of the seismic resilience of the TTLS under MSAS sequences is conducted. The framework enables provide decision-making in both pre-earthquake (e.g., structural design) and post-earthquake scenarios (e.g., recovery time), effectively mitigating the adverse consequences of sequential earthquakes.

2.1. Fragility analysis method

Fragility analysis is a vital method in the seismic resilience assessment, which establishes the relationship between structural demand and capacity and presents the conditional probability of reaching or exceeding a predefined damage state for a given earthquake intensity [39,40]. Based on the physics-based fragility method [41,42], it is assumed that the seismic capacity of an *EDP* follows a log-normal distribution and the fragility function can be expressed as following

$$P(S_D \geq S_C|IM) = 1 - \left(\frac{\ln(S_C) - \ln(S_D)}{\sqrt{\beta_{D|IM}^2 + \beta_C^2}} \right) \tag{1}$$

where $\Phi(\cdot)$ is the cumulative normal distribution function; S_C and S_D represent the median values of the structural capacity and demand, respectively; $\beta_{D|IM}$ and β_C are the logarithmic standard deviations of the structural demand and capacity, respectively. Probabilistic demand model is an effective tool to describe the relationship between *EDP* and *IM* and the mathematic expression is assumed to be an exponential distribution:

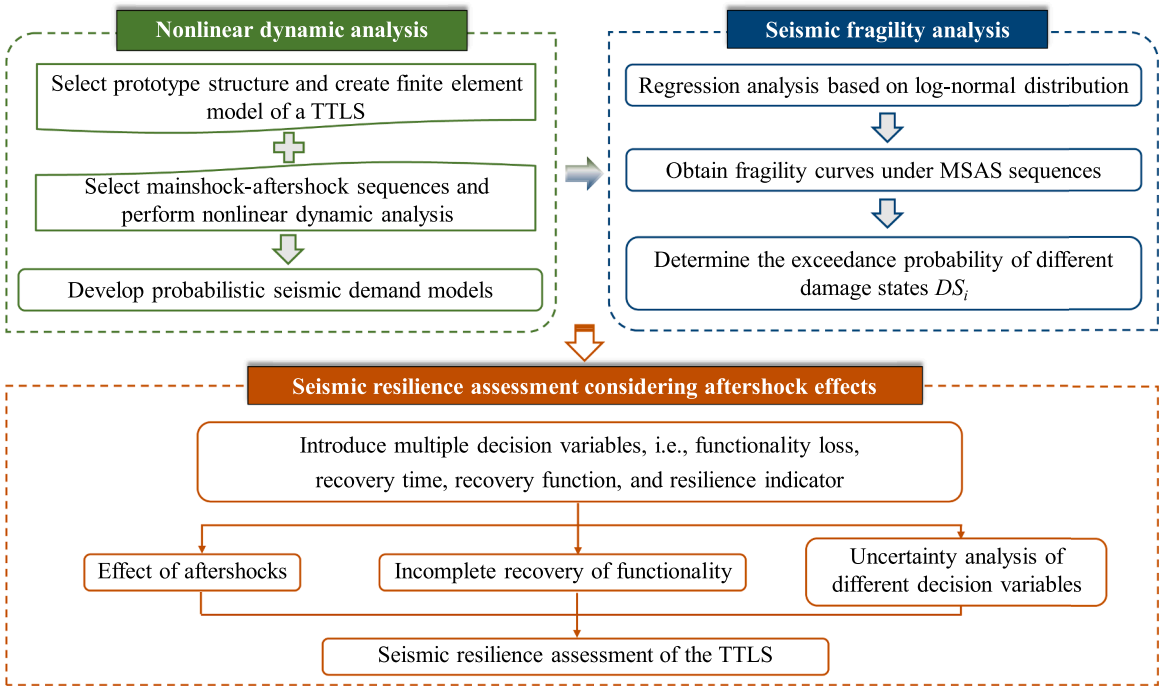


Fig. 1. Seismic resilience assessment framework considering aftershocks effect.

$$S_D = b \cdot IM^a \tag{2}$$

The above equation is generally rewritten in the form of the logarithm to preform linear regression analysis, which can be expressed as

$$\ln(S_D) = \ln(b) + a\ln(IM) \tag{3}$$

where a and b are regression coefficients that can be determined by linear fitting of the *EDP* data. $\beta_{D|IM}$ can be appropriately calculated as

$$\beta_{D|IM} \cong \sqrt{\frac{\sum (\ln(d_i) - \ln(S_D))^2}{N-2}} \tag{4}$$

where d_i is the structural responses of the TTLS under the ith earthquake; and N is the number of total simulations. β_C is considered as a constant of 0.25 [43].

In this study, we select the inter-segment displacement ratio (*ISDR*) as the *EDP* of the TTLS [44], while the peak ground acceleration (*PGA*) is used as the *IM*. MSAS sequences are scaled incrementally to obtain the sufficient data to construct fragility curves. A range of *IM* (0.2 g ~ 1.0 g) is scaled in the nonlinear dynamic analysis to capture the complete range of structural behaviors from elasticity to failure. There are four damage states (DS_i) that are categorized as follows: minor damage (DS_1), moderate damage (DS_2), major damage (DS_3) and collapse (DS_4). In order to quantify these damage states using the selected demand parameter *ISDR*, the threshold value of DS_4 is first selected as 4.0%, which aligns with the recommended threshold for the "complete damage" of the steel moment frames in HAZUS [45]. The threshold values for other damage states (DS_1, DS_2 and DS_3) are then obtained by multiplying the threshold value of DS_4 by different scaling factors (0.1, 0.2 and 0.4) [46]. Consequently, the threshold values of 0.4%, 0.8%, 1.6% and 4.0% for the different DS_i are adopted to establish the probabilistic demand model of the TTLS in the present study.

Identifying the probability of a structure being in any DS_i is crucial to estimate the functionality loss and recovery time of the TTLS [47]. Therefore, the cumulative probability of a fragility curve should be converted into a discrete probability, which can be expressed as following

$$P(ds = DS_i|IM) = \begin{cases} 1 - P(ds \geq DS_{i+1}|IM) & i = 0 \\ P(ds \geq DS_i|IM) - P(ds \geq DS_{i+1}|IM) & i = 1,2,3 \\ P(ds \geq DS_i|IM) & i = 4 \end{cases} \tag{5}$$

where $P(ds = DS_i|IM)$ is the conditional probability of the TTLS being in DS_i; DS_0 represents no damage; and ds is the intended damage state.

2.2. Seismic resilience assessment method

2.2.1. Resilience indicators

Seismic resilience refers to the ability of structures to maintain or recover their desired functionality after experiencing a strong earthquake over a period defined as the recovery time ($T_{RE|IM}$), which combines several variables (functionality loss, recovery time, etc.) and is an effective tool to assess the seismic performance of structures [48,49]. The resilience indicator R is generally calculated by the area below the functionality function $Q(t)$ over the time period from t_{OE} to $T_{RE|IM}$, as shown in Fig. 2. The horizontal axis represents the duration of post-earthquake recovery process and the vertical axis is the normalized

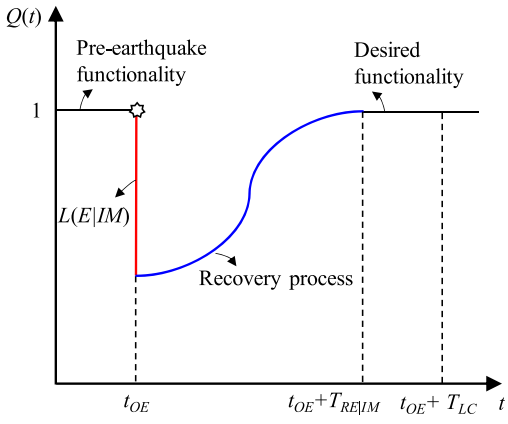


Fig. 2. Schematic diagram of the seismic resilience.

functionality ranging from 0 to 1. A value of R approaching 1 indicates a more resilient structure. When a structure is subjected to a single mainshock (see Fig. 2), R can be calculated as

$$R = \int_{t_{OE}}^{t_{OE}+T_{RE|IM}} \frac{1}{T_{RE|IM}} Q(t)dt \tag{6}$$

where R is the resilience indicator; t_{OE} is the moment at which an earthquake occurs; $T_{RE|IM}$ is the recovery time; and $Q(t)$ is a dimensionless functional function [50], and can be calculated as

$$Q(t) = 1 - L(E|IM) \times [H(t - t_{OE}) - H(t - (t_{OE} + T_{RE|IM}))] \times f_{rec}(t) \tag{7}$$

$$H(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} \tag{8}$$

where $H(x)$ is the Heaviside step function and can be expressed as Eq. (8); and $L(E|IM)$ and $f_{rec}(t)$ are the functionality loss and recovery function respectively, which will be described in detail below.

Importantly, it should be noted that various recovery processes with different recovery times $T_{RE|IM}$ can yield the same resilience indicator R. $T_{RE|IM}$ is a stochastic variable that is susceptible to several uncertain factors, such as seismic intensities, damage states and available resources for preparedness. Thus, the calculated R using Eq. (6) may exhibit significant uncertainties. To address this concern, Cimellaro et al. [51] suggested replacing $T_{RE|IM}$ with a customized control time T_{LC}, and R can be calculated using Eq. (9) instead. The control time T_{LC} can be chosen as the maximum recovery time, aiming to reduce the uncertainty associated with resilience indicators R. However, this modification fails to distinguish the resilience associated with different recovery functions (such as linear model and trigonometric model). In this respect, this paper draws on the suggestion of Sharma et al. [52] and introduces two parameters, the center of resilience ρ_Q and the resilience bandwidth χ_Q. The former combines the residual functionality and the corresponding recovery time, while the χ_Q signifies the degree of the spread of the recovery process [53]. ρ_Q and χ_Q can be calculated by Eqs. (10) and (11), respectively

$$R = \int_{t_{OE}}^{t_{OE}+T_{LC}} \frac{1}{T_{LC}} Q(t)dt \tag{9}$$

$$\rho_Q = \frac{\int_0^{T_{LC}} t\mathrm{d}Q(t)}{\int_0^{T_{LC}} \mathrm{d}Q(t)} \tag{10}$$

$$\chi_Q = \sqrt{\frac{\int_0^{T_{LC}} (t-\rho_Q)^2 \mathrm{d}Q(t)}{\int_0^{T_{LC}} \mathrm{d}Q(t)}} \tag{11}$$

In Eqs. (10) and (11), the value of t_{OE} is defined as 0. ρ_Q approximately represents the time at which $Q(t)$ reaches half of its value at $t = T_{LC}$. A small χ_Q value describes a rapid and concentrated functionality recovery around ρ_Q, while a large χ_Q value represents a relatively spread recovery process over time. Therefore, in assessing the post-earthquake recovery capabilities of the TTLS, multiple resilience indicators (R, ρ_Q and χ_Q) are employed. An excellent seismic resilience of the TTLS is demonstrated when it achieves higher R values along with lower ρ_Q and χ_Q values.

2.2.2. Influence of MSAS sequences

Previous research [54,55] has indicated that aftershocks often result in additional damage to structures that have already been weakened by the mainshock. These studies have widely agreed that the intervals between mainshocks and aftershocks are typically so short that there is insufficient time to develop recovery strategies and carry out restoration work. However, in case of transmission tower-line systems, the urgency to achieve post-earthquake functional recovery is paramount, as it plays a crucial role in restoring other electricity-dependent infrastructures within the shortest possible time. Consequently, retrofitting or maintenance measures may have been undertaken to partially or fully recover the functionality of mainshock-damaged TTLSs before the subsequent aftershocks occur. Nonetheless, relevant research examining the influence of incomplete recovery on seismic resilience of structures remains scarce. In view of this, this paper investigates three scenarios: $t_{gap} \approx 0$, $0 < t_{gap} < T_{RE|IM,MS}$, and $t_{gap} \geq T_{RE|IM,MS}$, as shown in Fig. 3. Here, t_{gap} represents the time gap between the mainshock and the subsequent aftershock; and $T_{RE|IM,MS}$ is the recovery time for the functionality of mainshock-damaged structures. Notably, this paper selects a single aftershock with the largest *PGA* to form a sequential earthquake along with its corresponding mainshock, which is considered acceptable as the additional damage to structures caused by multiple identical earthquakes only varies slightly from that of a single similar earthquake [56]. The same method of constructing MSAS sequences has been used in previous studies to strike a balance between efficiency and accuracy [29, 57].

Furthermore, it should be noted that the investigation of these three recovery scenarios aims to illustrate the potential recovery paths of post-disaster functionality in engineering structures subjected to chained or sequential hazards. The objective is also to analyze the influence of incomplete recovery on the resilience indicator of structures. Although this paper focuses on mainshock-aftershock sequences as an illustrative example, it is important to acknowledge that other multi-hazard situations may be more suitable for demonstrating different recovery scenarios.

(1) $t_{gap} \approx 0$

If there is not enough time to carry out maintenance on structures damaged by the mainshock before subsequent aftershocks occur, the resulting additional damage to the already weakened structures is illustrated in Fig. 3(a). In this particular scenario, the seismic resilience indicator of structures subjected to MSAS sequences can be calculated as

$$R = \int_{t_{OE}}^{t_{OE}+T_{LC}} \frac{1}{T_{LC}} Q_{MSAS}(t)dt \tag{12}$$

where $T_{RE|IM,MSAS}$ and $Q_{MSAS}(t)$ are the control time and functionality function of structures under MSAS sequences respectively; and $L_{MS}(E|IM)$ and $L_{AS}(E|IM)$ are the functionality losses under mainshocks and aftershocks respectively

(2) $0 < t_{gap} < T_{RE|IM,MS}$

When a mainshock-damaged structure undergoes a maintenance period of t_{gap} before the occurrence of aftershocks, its functionality is partially recovered but not to the desired level. The subsequent aftershocks will further diminish the structural functionality, which will be fully recovered after a maintenance period of $T_{RE|IM,MSAS}$. Fig. 3(b) presents the recovery process of the structural functionality, and the seismic resilience indicator in this specific scenario can be expressed as

$$R = \frac{1}{T_{LC}} \left(\int_{t_{OE}}^{t_{OE}+t_{gap}} Q_{MS}(t)\mathrm{d}t + \int_{t_{OE}+t_{gap}}^{t_{OE}+T_{LC}} Q_{MSAS}(t)\mathrm{d}t \right) \tag{13}$$

where $Q_{MS}(t)$ is the functionality function of structures under mainshocks.

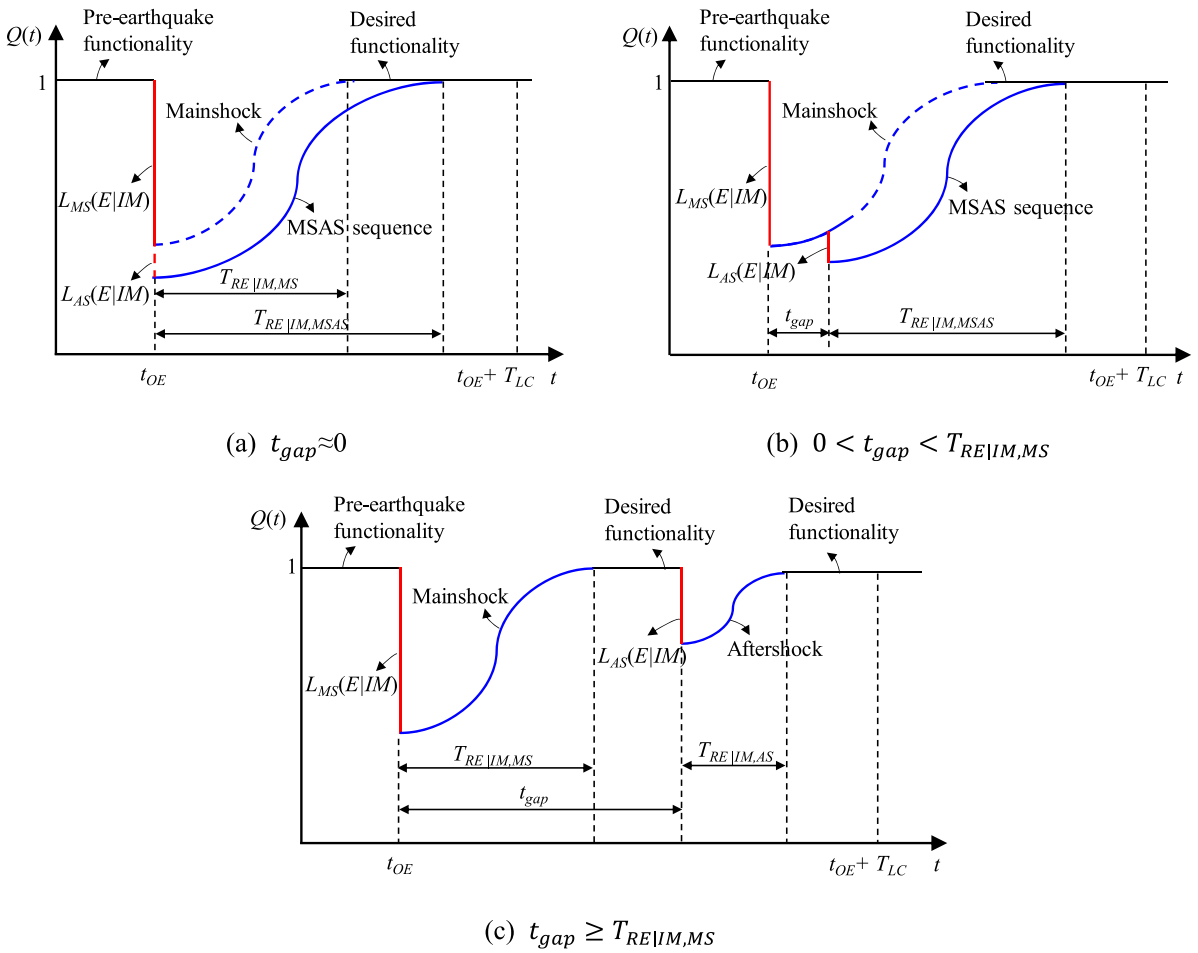


Fig. 3. The effect of different t_{gap} on the seismic resilience.

(3) $t_{gap} \geq T_{RE|IM,MS}$

Additional maintenance work must be performed to address the secondary damage caused by subsequent aftershocks since the time gap between the mainshock and aftershock is sufficient for the mainshock-damaged structure to regain its desired functionality, as shown in Fig. 3(c). In this scenario, the effects of the mainshock and aftershock are relatively independent, and the seismic resilience indicator can be expressed as

$$R = \frac{1}{T_{LC}}\left(\int_{t_{OE}}^{t_{OE}+t_{gap}} Q_{MS}(t)\mathrm{d}t + \int_{t_{OE}+t_{gap}}^{t_{OE}+T_{LC}} Q_{AS}(t)\mathrm{d}t\right) \tag{14}$$

where $Q_{AS}(t)$ and $T_{RE|IM,AS}$ are the functionality function and recovery time of structures under aftershocks, respectively.

2.2.3. Determination of the decision variables

(1) Functionality loss

Determining the structural functionality loss provides a quantitative basis for making the post-earthquake management decision regarding structures. This loss is commonly described in terms of economic losses [58]. Although casualty loss also is an important aspect of functionality loss, it is important to note that damage to TTLSs during earthquakes does not directly result in casualties, as TTLSs are typically constructed in remote mountainous or suburban areas with sparse population. In this study, the casualty loss is not considered in the seismic resilience assessment of the TTLS. The seismic functionality loss, denoted as $L(E|IM)$, mainly consists of direct loss $L_{dr}(E|IM)$ and indirect loss $L_{in}(E|IM)$, and can be expressed as

$$L(E|IM) = L_{dr}(E|IM) + L_{in}(E|IM) \tag{15}$$

$$L_{dr}(E|IM) = \sum_{i=1}^{n} P(ds = DS_i|IM)\cdot L_i \tag{16}$$

$$L_{in}(E|IM) = L_{dr}\cdot L_{dr}(E|IM) \tag{17}$$

where $L_{dr}(E|IM)$ and $L_{in}(E|IM)$ are the direct loss and indirect loss, respectively; L_i is the loss ratio; and L_{dr} is the loss coefficient.

In order to calculate the direct loss, Eq. (16) introduces a dimensionless parameter, loss ratio L_i, which is then multiplied by the probability of being in corresponding damage states. L_i is defined as the ratio of maintenance cost to reconstruction cost for a certain damage state of structures. According to *Post-earthquake field works-Part 4: Assessment of direct loss* (GB/T18208.4-2011) [59], The lower limit, mode, and upper limit values of L_i for different DS_i of TTLSs are listed in Table 1. It is assumed that L_i follows a triangular distribution [60], and the medians and variation ranges of L_i used subsequently to analyze the uncertainty

Table 1
L_i for different DS_i of TTLSs (%).

	DS_1	DS_2	DS_3	DS_4
Lower limit	6	16	36	56
Mode	11	26	46	68
Upper limit	15	35	55	80

in seismic resilience of TTLSs.

As for indirect loss $L_{in}(E|IM)$, it can be calculated by multiplying the loss coefficient L_{dr} by direct loss $L_{dr}(E|IM)$, as shown in Eq. (17).

(2) Recovery time

Recovery time is commonly defined as the overall duration required for a damaged structure to regain its desired functionality after an earthquake, which is influenced by various factors such as the structure's location, damage level, and importance. Given that the TTLS selected herein is one of the components of the electric power system, the medians for the recovery time $T_{RE|IM,i}$ at different DS_i can be referred to the recommended values in *HAZUS-MH MR4 technical manual* [61]. These values represent the time required for achieving 100% functional recovery. Table 2 lists the medians and standard deviations of $T_{RE|IM,i}$ for the TTLS at different DS_i. It is assumed that $T_{RE|IM,i}$ follows a normal distribution [61,62] and the uncertainty associated with $T_{RE|IM,i}$ is subsequently considered in seismic resilience of TTLSs. Likewise, it is assumed that the recovery time $T_{RE|IM}$ is solely dependent on the damage level of structures after earthquakes and can be calculated as following

$$T_{RE|IM} = \sum_{i=1}^{n} T_{RE|IM,i} \cdot P(ds = DS_i|IM) \tag{18}$$

where $T_{RE|IM,i}$ is the recovery time for a certain damage state DS_i of the TTLS.

(3) Recovery function

As mentioned previously, the post-earthquake recovery path of functionality is quite complicated, as it requires a comprehensive consideration of the uncertainties in social and economic dimensions. In the post-earthquake decision-making process, three commonly recovery models are employed to simulate the recovery process of post-earthquake functionality: linear, trigonometric, and exponential models [62]. The selection of a certain form of recovery model depends on social response and resource preparation. In particular, the linear model is adopted with a lack of social and resource response; the triangular model can be used when response is slow initially, but recovery can be achieved swiftly at the later stage; and the exponential model represents a scenario where structures respond rapidly initially, followed by a gradual reduction in recovery rate. Notably, these functional models are only at the conceptual level while do not precisely reflect the actual recovery process. Fig. 4 illustrates these recovery models, and the corresponding expressions can be expressed as

$$f_{rec}(t) = \begin{cases} a \times \left[\dfrac{t - t_{OE}}{T_{RE|IM}}\right] + b & \text{Linear} \\ \dfrac{c}{2} \times \left[1 + cos\pi d\left(\dfrac{t - t_{OE}}{T_{RE|IM}}\right)\right] & \text{Trigonometric} \\ e \times \exp\left[(-f) \times \dfrac{t - t_{OE}}{T_{RE|IM}}\right] & \text{Exponential} \end{cases} \tag{19}$$

where a, b, c, d, e and f are empirical parameters.

Table 2
$T_{RE|IM,i}$ for different DS_i of TTLSs (days).

	DS_1	DS_2	DS_3	DS_4
Median	0.5	1.5	4.5	10
Standard deviation	0.2	0.5	1.5	3

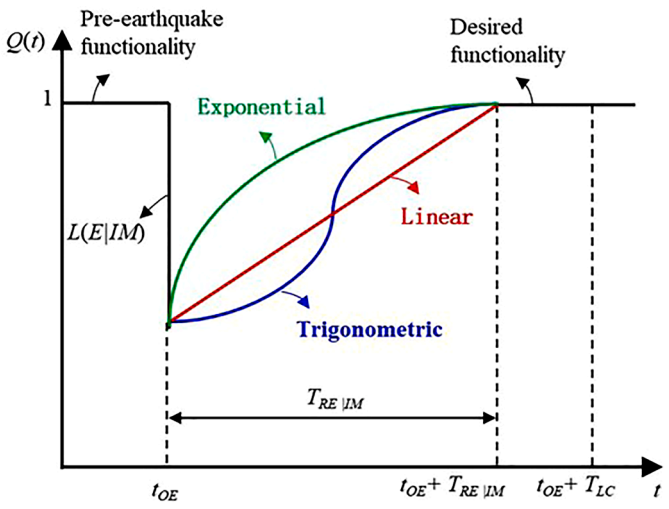


Fig. 4. Different recovery functions.

3. FE model and MSAS sequences

3.1. Development of FE model

The developed framework is applied to assess the seismic resilience of a 220 kV electricity transmission system under MSAS sequences. The prototype is designed in accordance with the current design specifications, which require a seismic fortification level of 8° of and a corresponding *PGA* of 0.2 g [63]. The analyzed subsystem composes of three suspension-type transmission tower (referred to as Towers 1–3) and four spans of transmission lines (namely Spans 1–4). These transmission towers are fabricated using Q420-, Q345- and Q235-type steel equal-leg angles, with a uniform design featuring a height of 46 m, a base dimension of 7.14 m × 7.14 m and a self-weight of 10.3 t The rationality of the design of the TTLS has been previously verified by comparisons between the full-scale test and numerical simulation [64]. Each transmission tower has four crossarms located at heights of 31.5, 37, 42.5, and 46 m. The topmost crossarm supports the ground lines, and the other crossarms support the conductors. The detailed properties of ground wires and conductors are listed in Table 3.

A FE model of the TTLS is established in the ABAQUS software, as shown in Fig. 5. All steel equal-leg angles are simulated by two-node three-dimensional beam elements (B31) and there are 1266 elements and 491 nodes in a transmission tower. Both transmission lines and insulators are modelled by using two-node three-dimensional truss elements (T3D2), with each span of the transmission line divided into 80 elements. The base of the transmission tower is fixed on the ground, neglecting soil-structure interactions. The far ends of the transmission lines at Spans 1 and 4 are designated as hinged. Geometric nonlinearities are considered by enabling the NLGEOM option in ABAQUS. For convenience of description, the transverse, longitudinal and vertical directions of the TTLS are denoted by X, Y and Z respectively.

Accurately defining the material properties of the steel angles is crucial for predicting the seismic performance of the TTLS. The steel

Table 3
Detailed properties of conductor and ground wires.

Description	Conductor line	Ground line
Type	ACSR/GA	OPGW24
Outside diameter (mm)	33.91	13.5
Weight (kg/m)	2.04	0.55
Modulus of elasticity (GPa)	62	110
Ultimate strength (kN)	140.5	110
Thermal expansion coefficient (1/ °C)	2.13E-5	1.5E-5

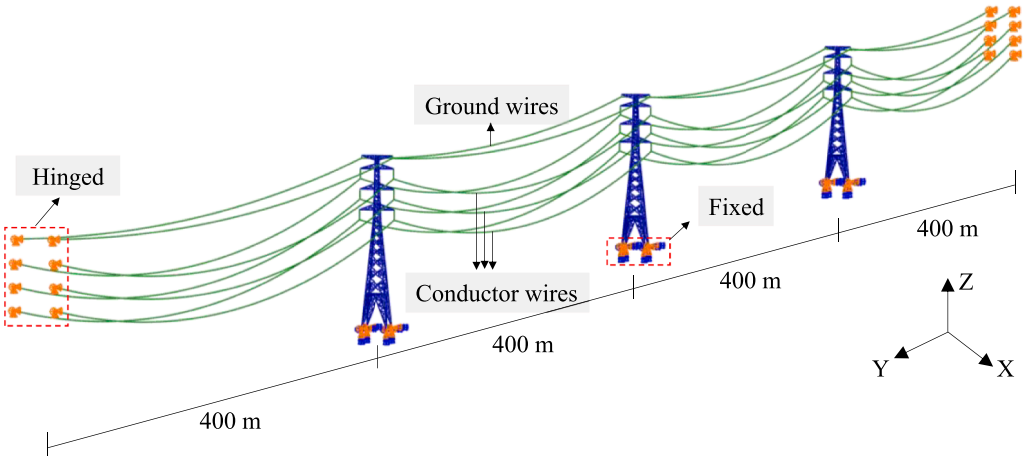


Fig. 5. FE model of the TTLS.

angles in the transmission tower have a large slenderness ratio and are prone to instability failure. Therefore, a nonlinear hysteretic model is introduced to simulate their complex behaviors, including strain hardening, buckling, and degradation of strength and stiffness. For conciseness, detailed information about the nonlinear hysteretic model is not included in this paper but can be found in [64,65]. The transmission lines are simulated by elastic tension-only material properties. What's more, fundamental natural frequencies of the TTLS in the X and Y directions are obtained by conducting eigenvalue analysis using Lanczos algorithm, yielding values of 0.468 Hz and 0.475 Hz, respectively.

3.2. Selection of MSAS sequences

Fifteen as-recorded MSAS sequences are selected from the NGA strong motion database of the Pacific Earthquake Engineering Research Center to assess the seismic resilience of the TTLS. In this study, the construction of the MSAS sequences adheres to the following criteria: (1) the acceleration response spectra of mainshocks should closely match the design spectrum, considering the site type and fortification requirement of the TTLS; (2) both mainshocks and aftershocks are selected from the same station of the same seismic event to eliminate spatial effects; (3) the magnitudes of the MSAS sequences are not less than M_w 5.0, and only one aftershock with the largest *PGA* is considered; (4) the mainshocks and aftershocks represent far-field ground motions. Detailed information regarding the selected mainshock-aftershock sequences is listed in Table 4. These sequential earthquakes are selected from different earthquake events including Northridge earthquake, Imperial Valley earthquake, etc. The occurrence time, magnitude M_w, fault distance R_{rup}, and shear wave velocity V_{s30} of these sequential earthquakes are also given. To ensure that the seismic response of the TTLS gradually reaches to a new equilibrium state before the occurrence of an aftershock, a time interval of 20 s with zero acceleration values is

Table 4
Detailed information of the selected mainshock-aftershock sequences.

No.	Events	Station	Type	Time	M_w	R_{rup} (km)	V_{s30} (m/s)
1	Northridge	LA-Baldwin Hills	*MS*	1994.1.17	6.69	18.36	545.66
			AS	1994.3.20	5.28	15.25	545.66
2	Northridge	LA-Chalon Rd	*MS*	1994.1.17	6.69	20.45	740.05
			AS	1994.3.20	5.28	17.01	740.05
3	Northridge	Malibu-Point Dume Sch	*MS*	1994.1.17	6.69	33.67	349.54
			AS	1994.3.20	5.28	37.33	349.54
4	Northridge	Moorpark-Fire Sta	*MS*	1994.1.17	6.69	24.76	341.58
			AS	1994.1.17	5.93	15.43	341.58
5	Northridge	LA-Wonderland Ave	*MS*	1994.1.17	6.69	20.29	1222.52
			AS	1994.3.20	5.28	17.14	1222.52
6	Imperial Valley	Brawley Airport	*MS*	1979.10.15	6.53	10.42	208.71
			AS	1979.10.15	5.01	25.32	208.71
7	Imperial Valley	Calexico Fire Statio	*MS*	1979.10.15	6.53	10.45	231.23
			AS	1979.10.15	5.01	13.32	231.23
8	Mammoth Lakes	Long Valley Dam	*MS*	1980.5.25	5.91	18.13	537.16
			AS	1980.5.25	5.7	14.38	537.16
9	Irpinia, Italy	Rionero In Vulture	*MS*	1980.11.23	6.9	30.07	574.88
			AS	1980.11.23	6.2	22.69	0.10
10	Whittier Narrows	Baldwin Park - N Holly	*MS*	1987.10.1	5.99	16.72	544.68
			AS	1987.10.4	5.27	16.03	544.68
11	Whittier Narrows	Burbank - N Buena Vist	*MS*	1987.10.1	5.99	26.34	320.57
			AS	1987.10.4	5.27	25.98	320.57
12	Whittier Narrows	LA - Fletcher Dr	*MS*	1987.10.1	5.99	39.45	270.54
			AS	1987.10.4	5.27	40.38	270.54
13	Whittier Narrows	Sunland - Mt Gleason Ave	*MS*	1987.10.1	5.99	30.37	402.16
			AS	1987.10.4	5.27	30.27	402.16
14	Chi-Chi	HWA056	*MS*	1999.9.20	7.62	35.9	511.3
			AS	1999.9.22	6.2	54.98	511.3
15	Chi-Chi	HWA058	*MS*	1999.9.20	7.62	45.77	529.54
			AS	1999.9.22	6.2	44.07	529.54

inserted between the mainshocks and aftershocks. Fig. 6 shows the acceleration spectra and mean spectrum of selected MSAS sequences. Fig. 7 gives an illustration of the selected MSAS sequences.

4. Seismic resilience assessment considering aftershocks

4.1. Fragility analysis

The scatter distribution of the ln(*ISDR*) of the TTLS under MSAS sequences is shown in Fig. 8, where the blue dashed lines represent the threshold values of different DS_i, corresponding to ln(*ISDR*) of -5.52, -4.83 -4.14 and -3.22 for DS_1, DS_2, DS_3 and DS_4, respectively. The red dots indicate the average values of the ln(*ISDR*) at a given ln(*PGA*). As shown in Fig. 8(a), *ISDR* increases with the increase of *PGA*. When *PGA* is less than 0.4 g, all scatter points are located below the blue dashed line of DS_3, indicating that the TTLS mainly suffers from minor and moderate damage. Once *PGA* exceeds 0.8 g, the average values of ln(*ISDR*) are located above the blue dashed line of DS_4, indicating that the TTLS is highly susceptible to collapse. Similar conclusions can be obtained in Fig. 8(b). However, at a given *PGA*, the scatter points for the TTLS under MSAS sequences are generally located above those for mainshocks. When *PGA* ranges from 0.2 to 0.5 g, the distribution of ln(*ISDR*) presents significant dispersion. Once *PGA* exceeds 0.7 g, the scatter points of the TTLS under MSAS sequences tend to cluster in the region of DS_4. This phenomenon can be attributed to the additional damage caused by subsequent aftershocks, which further increases the vulnerability of the already damaged TTLS to collapse following mainshocks.

Fig. 9(a) shows the fragility curves of the TTLS under sequential earthquakes. Some general conclusions can be drawn: the exceedance probability gradually increases with the increase of *PGA*, and the fragility curves shift towards the right as the damage states gradually process from DS_1 to DS_4. Compared to mainshocks, the fragility curves for each DS_i of the TTLS under MSAS sequences are located on the left, indicating a higher exceedance probability at a given *PGA* at a same damage state. Furthermore, the probability of the TTLS being in a specific DS_i is obtained by discretizing the cumulative probability, as shown in Fig. 9(b). It can be observed that for DS_2 and DS_4, the associated probabilities show a trend of first increasing and then decreasing as *PGA* increases. At a certain *PGA* value, the discrete probability curves for the TTLS under MSAS sequences intersect with those for mainshocks. For instance, Point A and Point B represent the intersection points for DS_2 and DS_4 respectively. Taking Point A as an example, when *PGA* is less than 0.34 g, the probability of the TTLS being in DS_2 under MSAS sequences is higher; otherwise, the probability is lower. This can be attributed to the additional damage caused by subsequent aftershocks, thereby increasing the probability of experiencing more severe damage states.

4.2. Analysis of the decision variables

(1) Analysis of functionality loss

The functionality losses $L(E|IM)$ of the TTLS under sequential earthquakes can be obtained by combining the discrete probability and loss ratio, as shown in Fig. 10. $L(E|IM)$ gradually increases as *PGA* increases. When *PGA* reaches 1.0 g, $L(E|IM)$ of the TTLS under mainshocks and MSAS sequences are 0.94 and 0.99 respectively, indicating that the repair cost for the damaged TTLS is close to the reconstruction cost. Fig. 10 also quantifies the comparison of $L(E|IM)$ between mainshocks and MSAS sequences using the difference ($L_{MSAS}(E|IM) - L_{MS}(E|IM)$). It can be observed that with the increase of *PGA*, the difference of the functionality loss increases initial and then decreases, reaching maximum values of 0.077 at a *PGA* of 0.7 g. This signifies that the effect of aftershocks becomes more significant when *PGA* is in the range of 0.6~0.7 g.

(2) Analysis of recovery time

The recovery time of the functionality of the damaged TTLS is shown in Fig. 11. An increase in *PGA* from 0.2 g to 1.0 g leads to longer recovery times of the TTLS under both mainshocks and MSAS sequences, ranging from 0.17 to 8.76 days and 0.25 to 9.51 days, respectively. Moreover, the difference ($T_{RE|IM,MSAS} - T_{RE|IM,MS}$) of the recovery time are demonstrated in Fig. 11. The trend in the difference of the recovery time aligns with the functionality loss of the TTLS, wherein it initially increases and then decreases with the increase of *PGA*. This phenomenon is due to the significant effect of subsequent aftershocks on the damage states of the TTLS when it suffers from minor or moderate damage after mainshocks. However, once *PGA* exceeds 0.8 g, the TTLS is likely to undergo major damage or collapse under the excitation of mainshocks, leading to a reduced effect of subsequent aftershocks.

(3) Analysis of recovery process

The comparisons between different recovery functions (i.e., linear, trigonometric, and exponential models) are shown in Fig. 12. As *PGA* increases, the starting points of the recovery processes gradually shift downward, while the end points move to the right. This indicates more

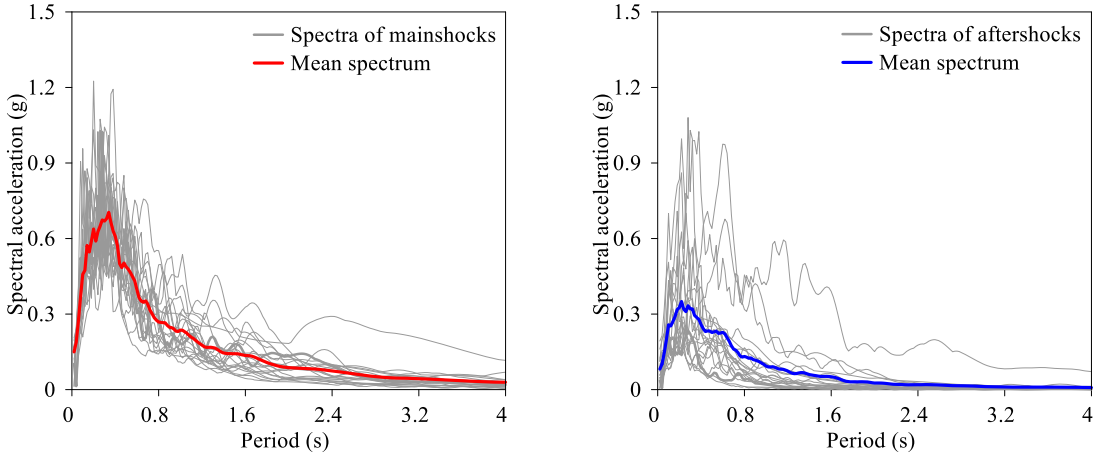


Fig. 6. Acceleration spectra and mean spectrum of selected MSAS sequences.

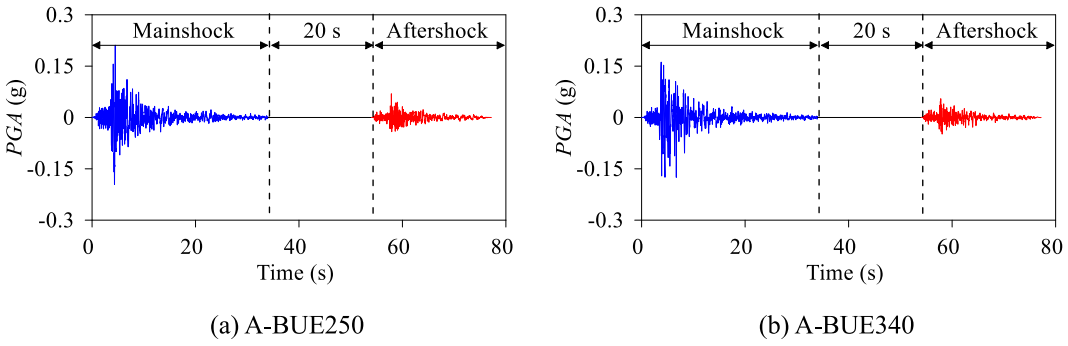


Fig. 7. Acceleration time histories of Whittier Narrows earthquake at Burbank - N Buena Vista.

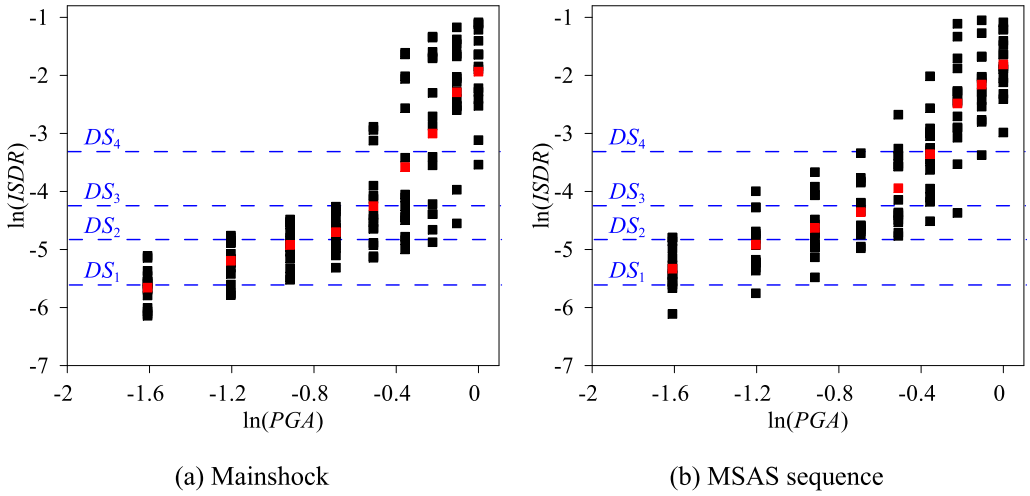


Fig. 8. Probabilistic seismic demand models of the TTLS.

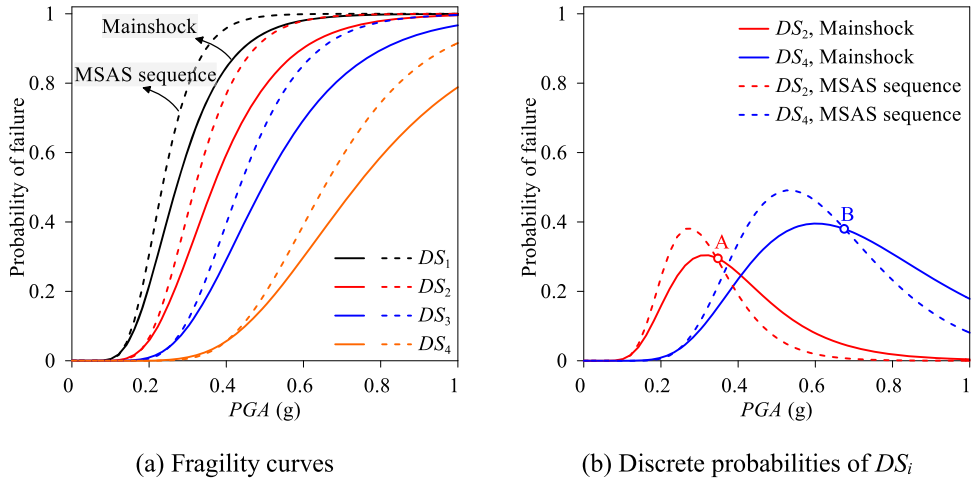


Fig. 9. Probability of failure of the TTLS.

severe functionality losses and longer recovery times. Furthermore, there are significant differences among the functional recovery processes of the TTLS predicted by the different recovery models. The post-earthquake functionality obtained from the linear model exhibits a linear increase over time. Conversely, the functional recovery predicted by the trigonometric model initially progresses slowly until it reaches $T_{RE|IM}/2$, after which it rapidly accelerates in the later stages of the recovery process, as shown in Fig. 12(b). As for the exponential model, the post-earthquake functionality increases rapidly during the initial stages of the recovery process, with more than 90% of functionalities recovered by the time $T_{RE|IM}/2$ is reached.

The effect of the time intervals t_{gap} between mainshocks and subsequent aftershocks on the functional recovery process of the TTLS using the trigonometric model at a *PGA* of 0.5 g, as shown in Fig. 13. As can be seen from Fig. 13(a), the functional recovery curve of the TTLS under MSAS sequences resides at the lower right of the recovery curve under mainshocks, indicating that the TTLS suffers more severe functionality losses and requires longer recovery times. For $0 < t_{gap} < T_{RE|IM,MS}$, the

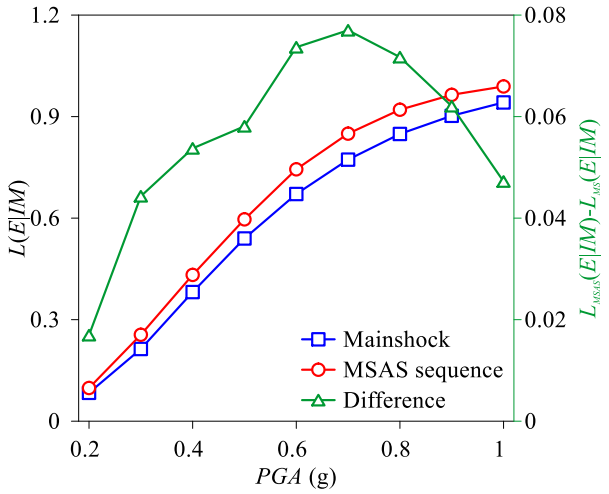


Fig. 10. Influence of aftershocks on $L(E|IM)$.

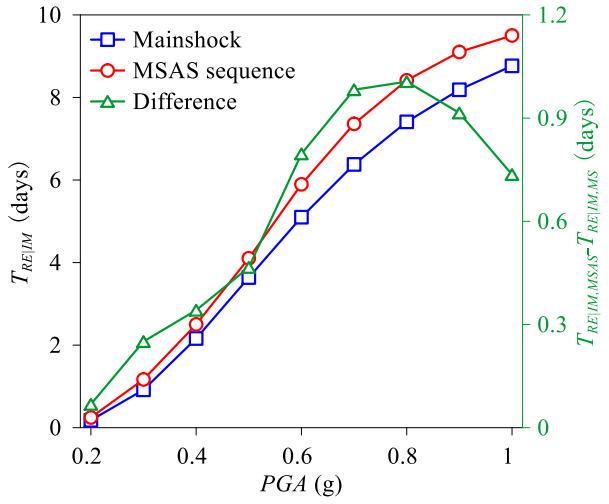


Fig. 11. Influence of aftershocks on T_{RE}.

TTLS that has been damaged by mainshocks can regain functionality of 0.33 to 0.43 after undergoing repairs for 1.34 days. However, the occurrence of subsequent aftershocks leads to a reduction in functionality. To restore the TTLS to its desired state, an additional 4.5 days of repairs are required. It can be seen from Fig. 13(c) that even after 5.1 days of repairs, the TTLS exhibits only temporarily recovered, reverting to a damaged state when subjected to subsequent aftershocks. Consequently, supplementary repair measures must be taken to reinstate its functionality.

(4) Analysis of resilience indicators

Fig. 14 compares R of the TTLS under sequential earthquakes. The R of the TTLS under MSAS sequences are smaller than those under mainshocks, indicating that the additional damage inflicted by subsequent aftershocks weakens the ability of the TTLS to recover its functionality. A comparison of R predicted by different recovery models is also presented. R gradually decreases with the increase of *PGA* for these recovery functions (i.e., linear, trigonometric, and exponential models). The exponential model exhibits a comparatively larger R compared to the other two recovery functions. Although the linear and trigonometric models present different recovery processes, the resulting curves of R mostly overlap, which is why it is difficult to distinguish the resilience associated with different recovery functions using Eq. (9).

In view of this, the resilience center ρ_Q and resilience bandwidth χ_Q calculated by Eqs. (10) and (11) respectively are used to illustrate the difference among the different recovery models. A comparison of ρ_Q is given in Fig. 15. It can be seen that for all recovery models, ρ_Q gradually increases as *PGA* increases, and ρ_Q of the TTLS under MSAS sequences are all higher than those under mainshocks, indicating that more time is needed to recover the post-earthquake functionality. In addition, the exponential model yields a relatively smaller ρ_Q compared to the other two recovery models. Specifically, when *PGA* reaches 1.0 g, ρ_Q decreases by 61.3%. However, the ρ_Q obtained by the linear and trigonometric models still show the same trend. A comparison of χ_Q is shown in Fig. 16. It can be seen that the exponential model still has the lowest χ_Q, indicating that this recovery model enables better recovery to the TTLS. However, the trigonometric model enables a relatively lower χ_Q compared to the linear model, indicating that a larger percentage of the recovery process can be completed within a shorter period of time around ρ_Q.

Considering different time intervals t_{gap} between mainshocks and subsequent aftershocks, a comparison of R of the TTLS under MSAS sequences is shown in Fig. 17. It can be seen that the trends of R obtained from the three recovery models are similar. When the TTLS is subjected to aftershocks after its mainshock-damaged functionality is fully recovered, R is maximum. Conversely, the minimum R is detected when no repairs are made to the damaged TTLS before the occurrence of subsequent aftershocks. For the time intervals within the range of $0 < t_{gap} < T_{RE|IM,MS}$ (taking $t_{gap} = T_{RE|IM,MS}/2$ as an example), the resulting R is similar to those observed when t_{gap} is approximately 0.

Fig. 18 investigates the effect of the incomplete recovery of the post-earthquake functionality on R for time intervals within the range of $0 < t_{gap} < T_{RE|IM,MS}$. For clear description, a ratio of the time interval to the post-mainshock recovery time ($t_{gap}/T_{RE|IM,MS}$) is introduced. A ratio of 0 indicates that the mainshock-damaged TTLS is not repaired before the occurrence of aftershocks, while a ratio of 1 denotes that the functionality of the mainshock-damaged TTLS is fully recovered before subsequent aftershocks occur. It is noteworthy that for the different recovery models, there is a similar trend observed in the variation of R with $t_{gap}/T_{RE|IM,MS}$. Specifically, R gradually increases with the increase of $t_{gap}/T_{RE|IM,MS}$. This implies that the more the functionality of the mainshock-damaged TTLS is restored, the more rapidly the TTLS recovers its functionality after subsequent aftershocks.

5. Uncertainty in seismic resilience

More importantly, the functionality loss and recovery time of the TTLS are not constant and can be affected by various uncertain factors such as structural damage levels and recovery strategies. Additionally, the post-recovery functionality of the TTLS may be lower than, equal to, or higher than the pre-earthquake level. In cases where the TTLS suffers to severe damage that is challenging to repair, its functionality may be lower than the desired level. Conversely, retrofitting or strengthening measures could result in a higher functionality than the pre-earthquake level. Hence, a comprehensively assessment of the seismic performance of the TTLS necessitates a reasonable consideration of the uncertainties associated with functionality loss, recovery time, and post-recovery functionality.

5.1. Uncertainty in functionality loss

Table 1 lists the medians and variation ranges of the loss ratio L_i for different DS_i of TTLSs, and the L_i can be assumed to follow a triangular distribution [60]. Therefore, for each specific *PGA*, 100 groups of L_i for different DS_i are generated using Monte-Carlo simulation. Subsequently, Eqs. (15)–(17) are adopted to calculate 100 groups of uncertain functionality loss and the effect of the functionality loss uncertainty on the

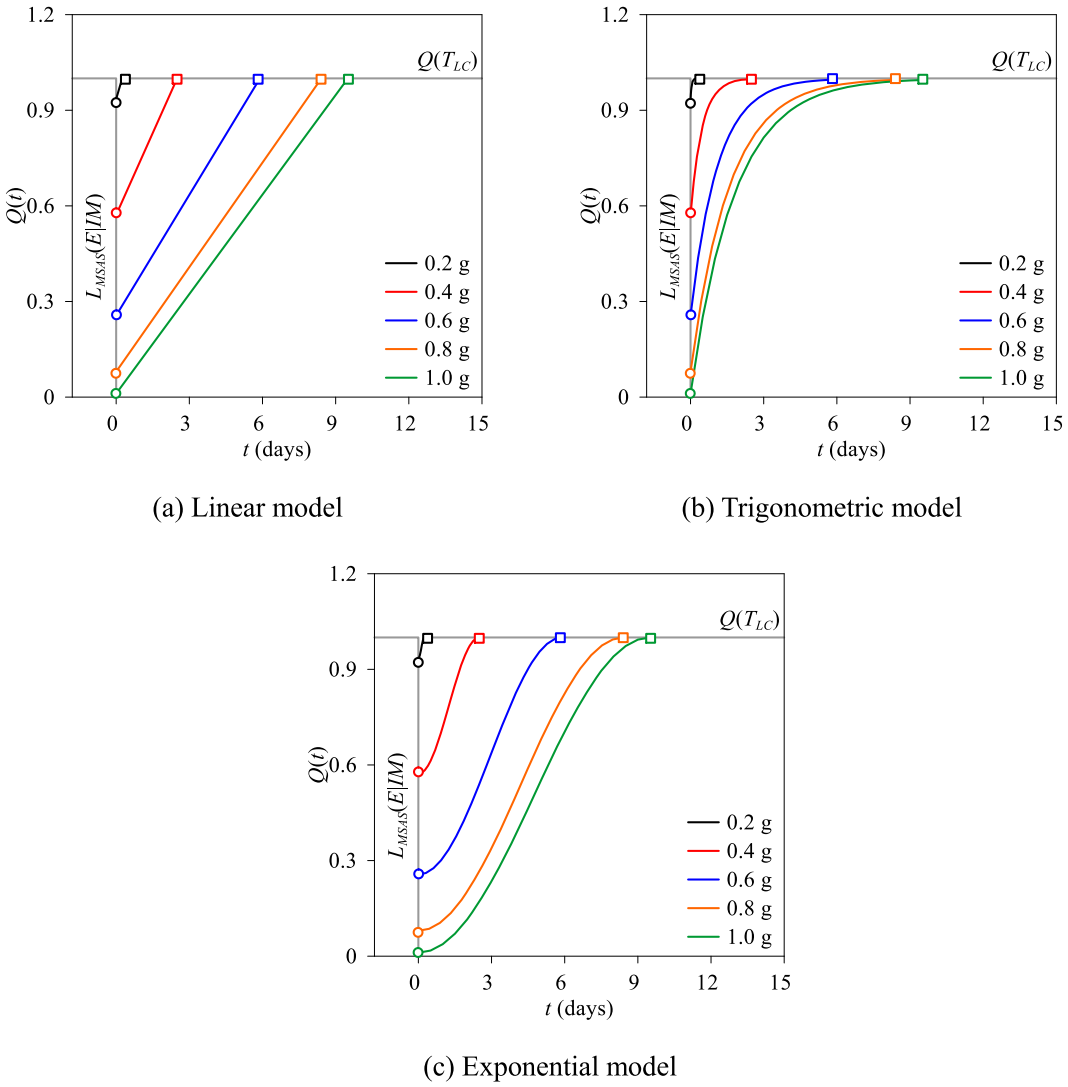


Fig. 12. Comparison of different recovery functions.

seismic resilience of the TTLS can be investigated. The results, predicted by the trigonometric model, are presented in Fig. 19.

It can be seen from Fig. 19(a) that the difference between the upper and lower limits of $L(E|IM)$ gradually increases as *PGA* increases from 0.2 to 0.8 g. When *PGA* exceeds 0.8 g, the upper limits of $L(E|IM)$ reach 1.0, indicating that the cost for recovering its post-earthquake functionality is equal to the reconstruction cost. What's more, the upper and lower limits as well as the 50% percentile values of $L(E|IM)$ for the TTLS under MSAS sequences are higher than those under mainshocks. This suggests that subsequent aftershocks result in additional losses. Fig. 19 (b) presents the recovery processes predicted by the trigonometric model at a *PGA* of 0.6 g. When $t = t_o$, the uncertainty of $L(E|IM)$ is most significant, with its effect decreasing gradually as the restoration progresses. The effect of the uncertainty in $L(E|IM)$ on R is shown in Fig. 19 (c). As *PGA* increases, R decreases gradually for all uncertain cases. When *PGA* is 1,0 g, the upper and lower limits of R under mainshocks are 0.56 and 0.64, respectively, with a difference of 0.08. Similarly, the upper and lower limit values of R under MSAS sequences are 0.52 and 0.59, respectively, with a difference of 0.07.

5.2. Uncertainty in recovery time

Table 2 lists the medians and standard deviations of recovery time $T_{RE|IM,i}$ for different DS_i of TTLSs, and the $T_{RE|IM,i}$ can be assumed to follow a normal distribution [61,62]. Similarly, 100 groups of $T_{RE|IM,i}$ for different DS_i are generated using Monte-Carlo simulation, and the corresponding uncertain values of $T_{RE|IM}$ are calculated using Eq. (18). Fig. 20 shows the effect of the uncertainty in $T_{RE|IM}$ on the seismic resilience of the TTLS.

Some general conclusions can be drawn: as *PGA* increases, $T_{RE|IM}$ gradually increases, while R decreases; the difference between the upper and lower limit values ($T_{RE|IM}$ and R) becomes significant; and the TTLS requires a longer $T_{RE|IM}$ to recover its functionality and has a lower R under MSAS sequences. However, the uncertainty of $T_{RE|IM}$ is more significant compared to that of $L(E|IM)$, and the distribution of these scatter points gradually widens with the increase of *PGA*. When *PGA* reaches 1.0 g, the upper and lower limit values of R for mainshocks are 0.36 and 0.83, respectively, with a difference of 0.47; for MSAS sequences, these values are 0.27 and 0.82, respectively, with a difference of 0.55. As shown in Fig. 20(b), due to the certainty of $L(E|IM)$, the recovery processes at the time of t_o start at the same point. As the restoration progresses, the recovery curves gradually shift to the right and show significant differences.

5.3. Uncertainty in both functionality loss and recovery time

Figs. 19 and 20 solely examine the effect of uncertainty in a single decision variable (functionality loss or recovery time) on the seismic

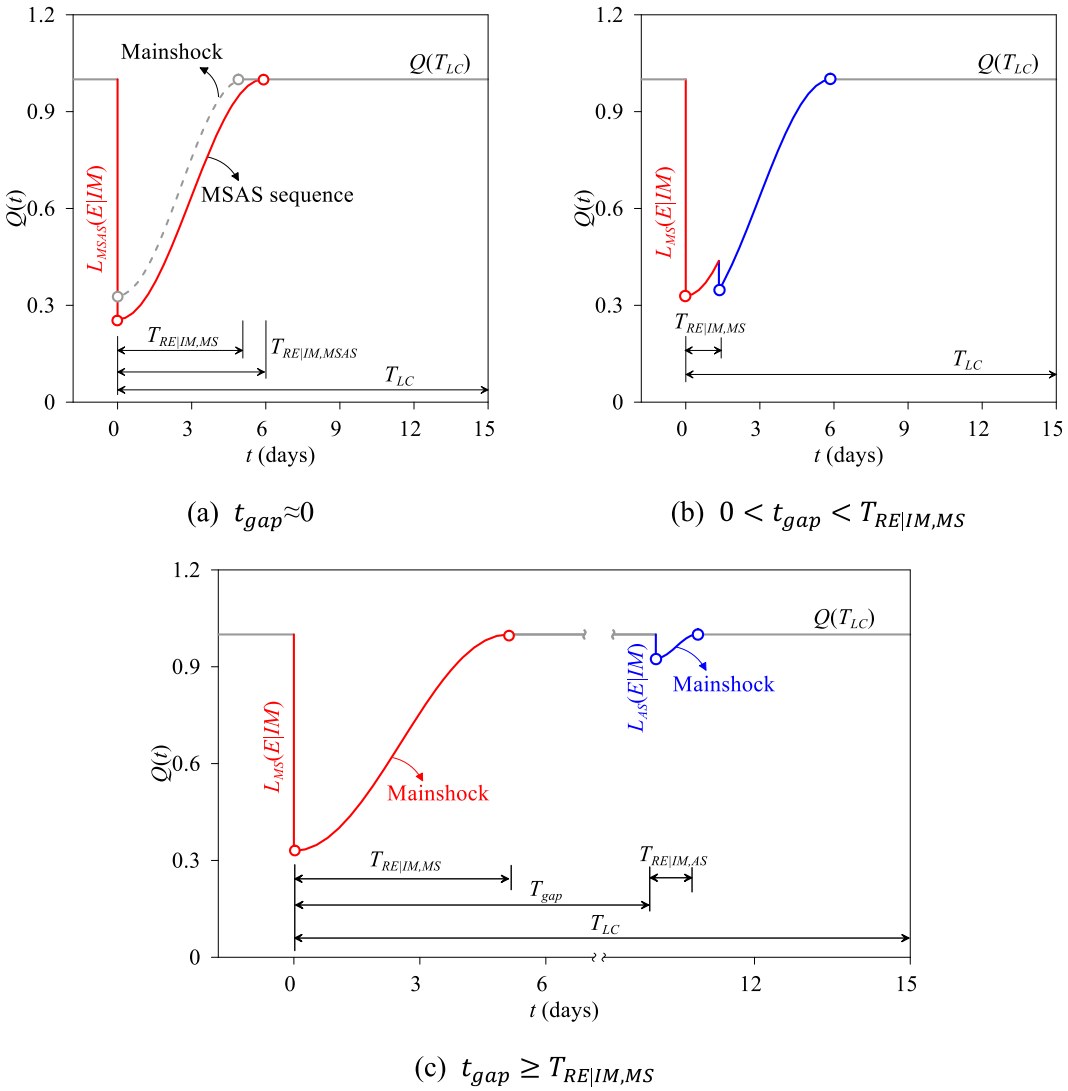


Fig. 13. Recovery process for different time intervals.

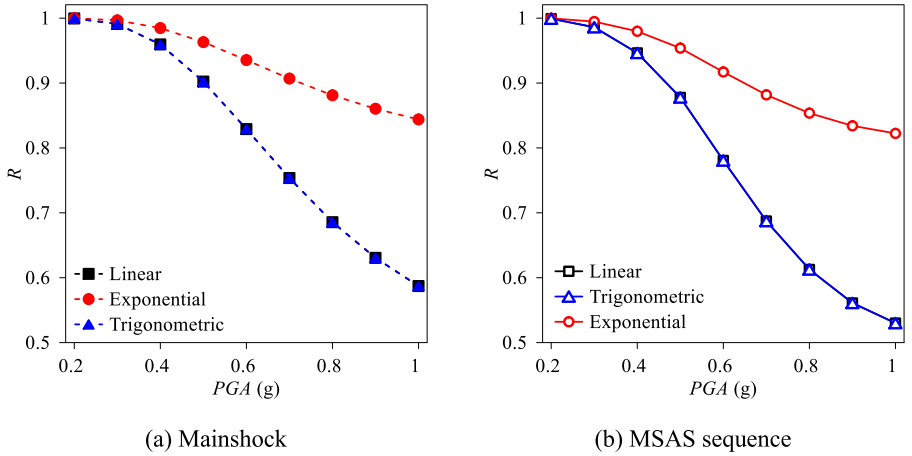


Fig. 14. Comparison of R.

resilience of the TTLS. However, both functionality loss and recovery time exert a significant influence on the recovery process and seismic resilience. Thus, it is essential to further consider the uncertainties associated with both functionality loss and recovery time. By combining the uncertainty samples of $L(E|IM)$ and $T_{RE|IM}$ generated in Sections 5.1 and 5.2, the seismic resilience of the TTLS under MSAS sequences can be obtained, as shown in Fig. 21. Due to the uncertainties in both $L(E|IM)$ and $T_{RE|IM}$, the scatter points in the recovery processes exhibit a more

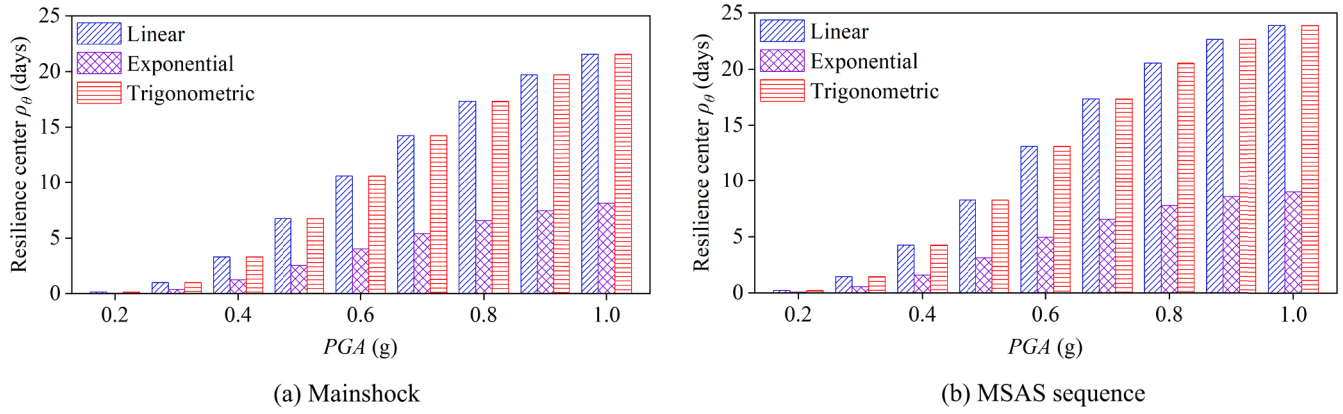


Fig. 15. Comparison of ρ_θ.

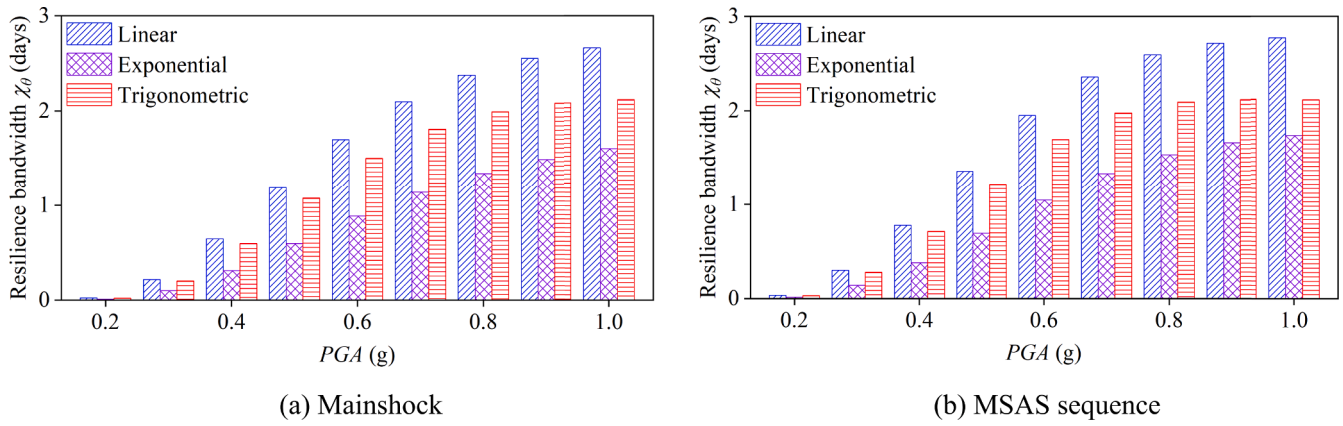


Fig. 16. Comparison of χ_θ.

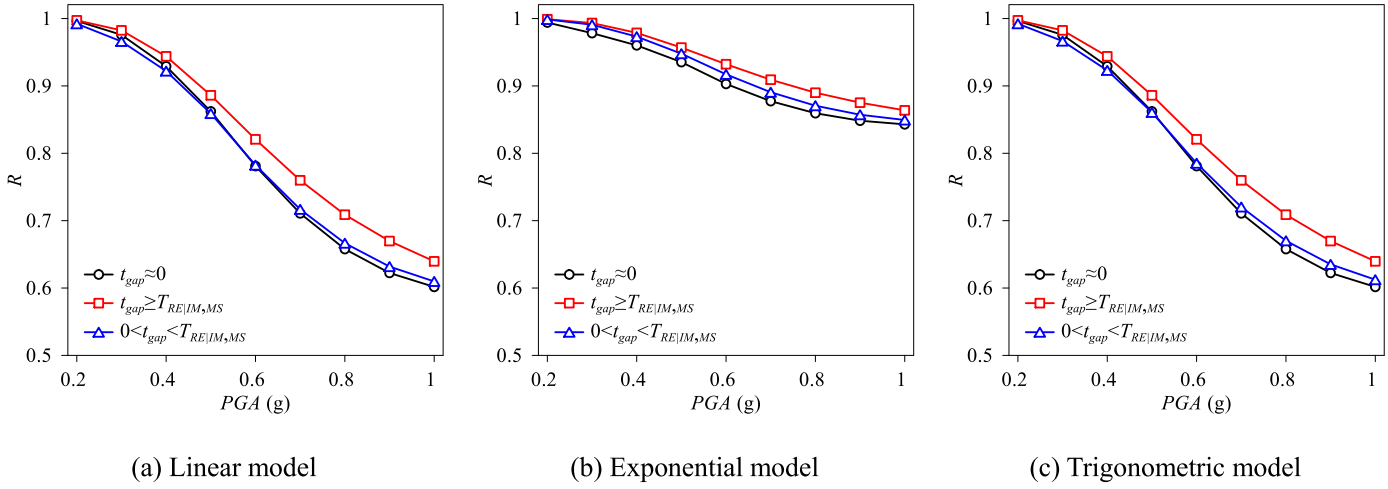
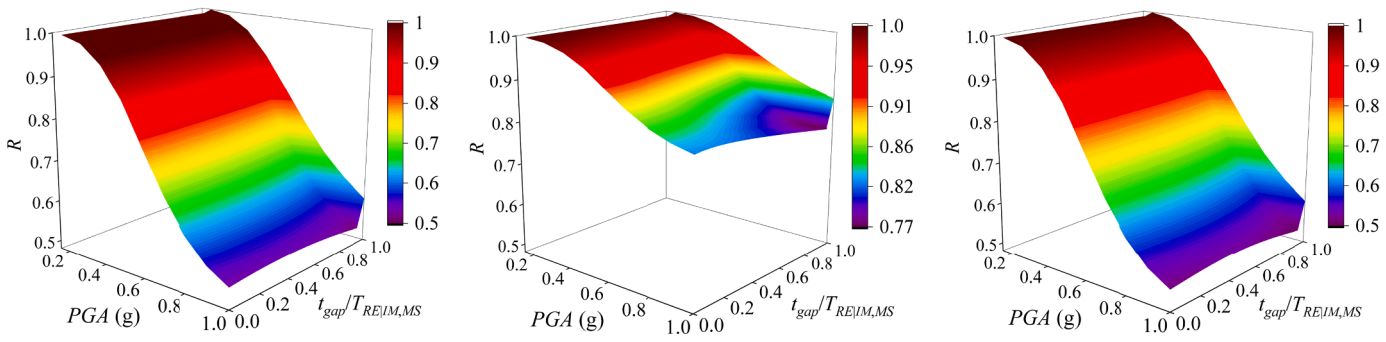


Fig. 17. Effect of different t_{gap} on R.

significant dispersion, similar to the distribution of the scatter points of R, as shown in Fig. 21(b). When PGA is 1.0 g, the upper and lower limit values of R for mainshocks are 0.33 and 0.86, respectively, with a difference of 0.53. For MSAS sequences, the corresponding values of R are 0.26 and 0.85, respectively, with a difference of 0.59. These values closely align with the results obtained solely considering the uncertainty of $T_{RE|IM}$, without accounting for the uncertainty of $L(E|IM)$. Consequently, the seismic resilience assessment of TTLSs should emphasize the uncertainty of $T_{RE|IM}$. In post-earthquake recovery decision-making, it is imperative to developed reasonable recovery strategies that facilitate prompt restoration and enhance the ability of TTLSs to withstand extreme disasters.

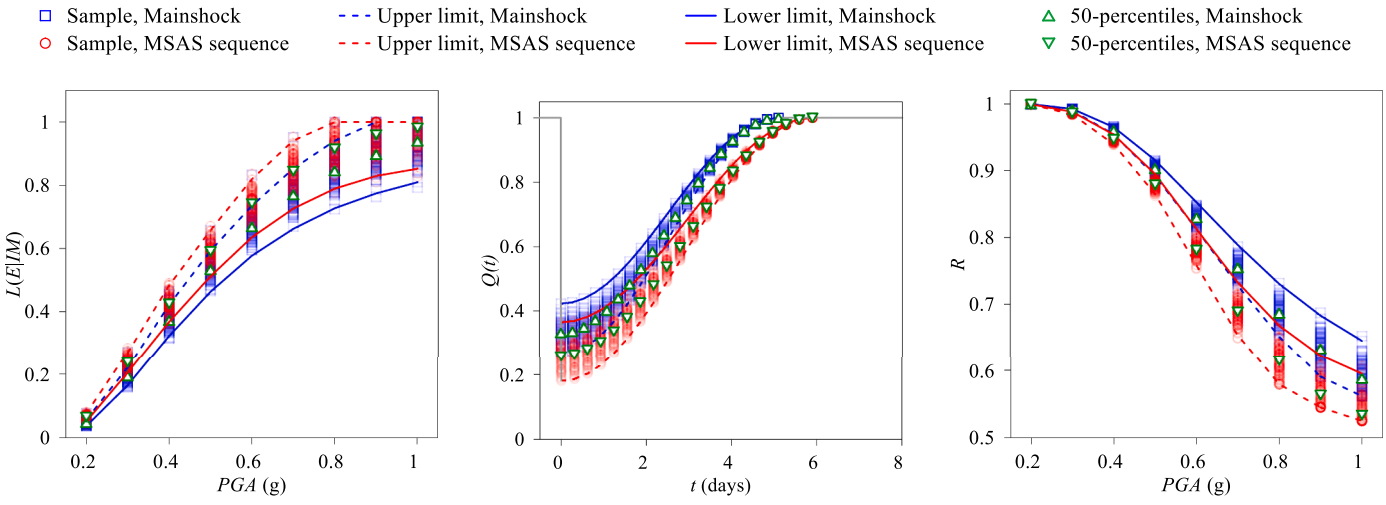
5.4. Uncertainty in post-recovery functionality

We select several cases of the post-recovery functionality $Q(T_{RE|IM})$ to be 0.8, 0.9, 1.0, 1.1 and 1.2, in order to investigate the influence of the uncertainty of $Q(T_{RE|IM})$ on the seismic resilience of the TTLS. Fig. 22



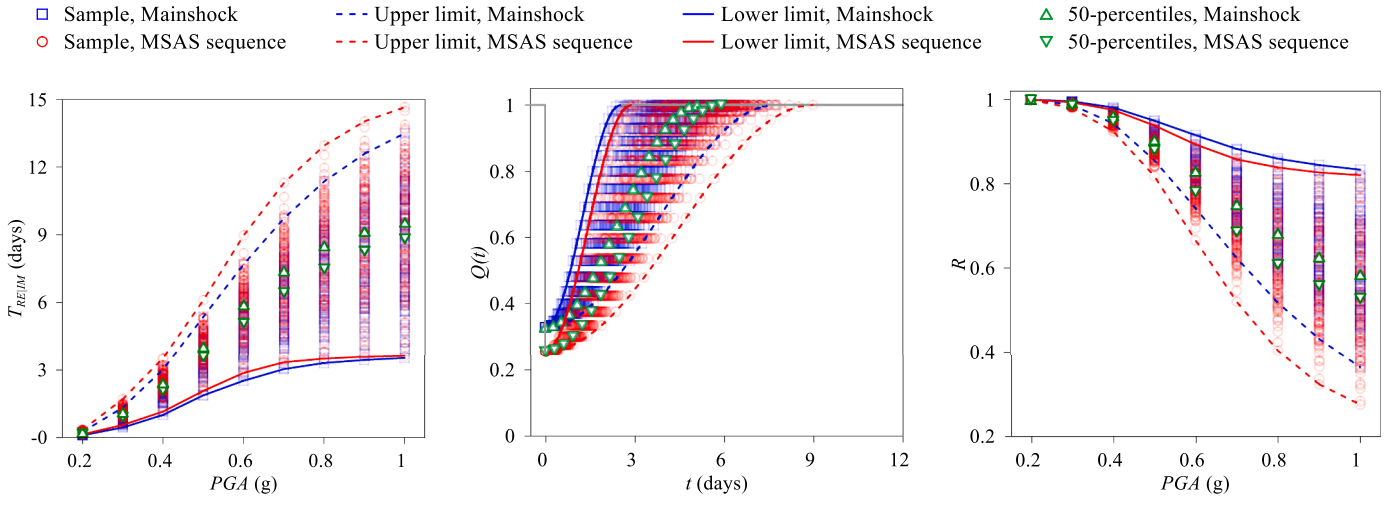
(a) Linear model (b) Exponential model (c) Trigonometric model

Fig. 18. Comparison of R for $0 < t_{gap} < T_{RE|IM,MS}$.



(a) Functionality loss (b) Recovery process (c) R

Fig. 19. Effect of uncertainty in $L(E|IM)$ on the seismic resilience of the TTLS.



(a) Recovery time (b) Recovery process (c) R

Fig. 20. Effect of uncertainty of $T_{RE|IM}$ on the seismic resilience of the TTLS.

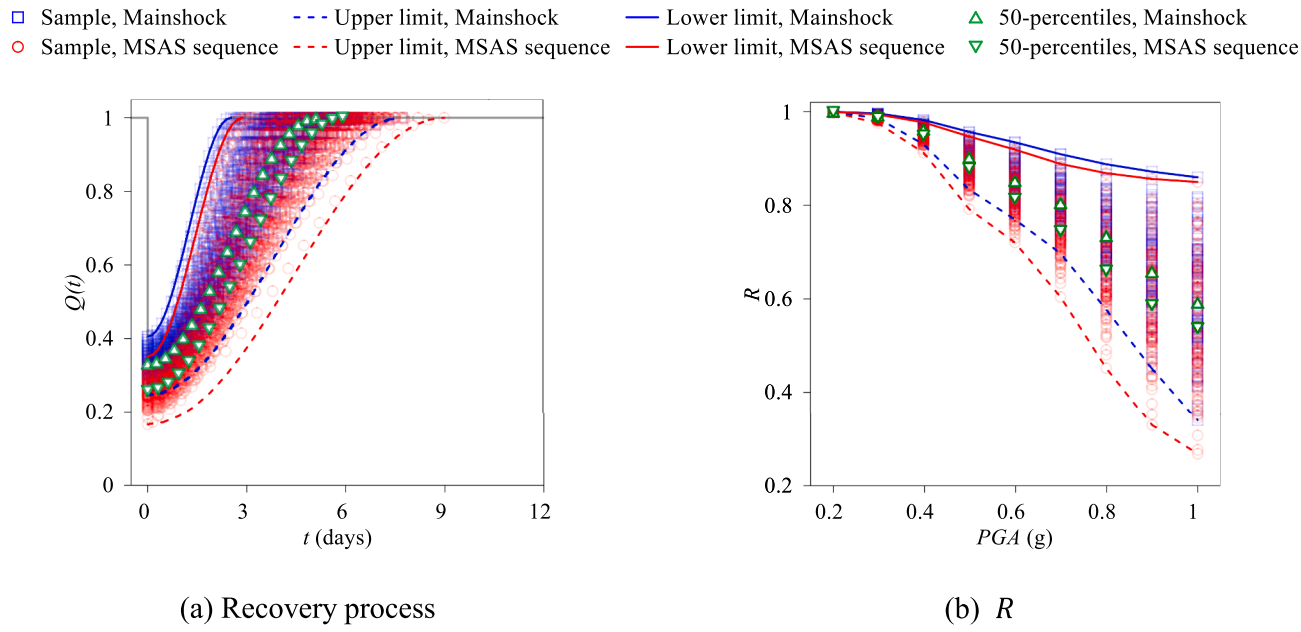


Fig. 21. Effect of uncertainty of both $T_{RE|IM}$ and $L(E|IM)$ on the seismic resilience.

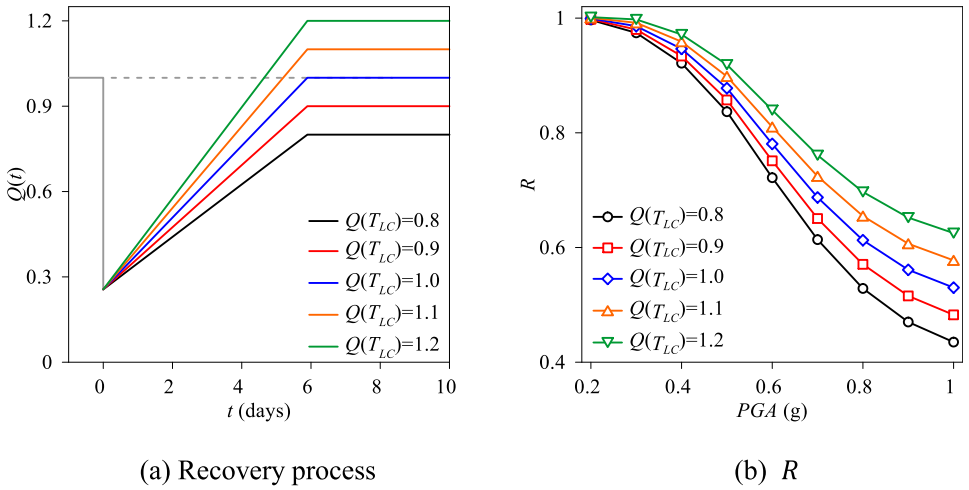


Fig. 22. Influence of different post-recovery functionality on the seismic resilience.

shows how varying $Q(T_{RE|IM})$ affecting the recovery process, as predicted by the linear model when *PGA* is 0.6 g. Different $Q(T_{RE|IM})$ have a significant influence on the recovery process of the TTLS. As $Q(T_{RE|IM})$ increases, a greater slope is observed meaning that more efficient responses are required to repair the damaged TTLS after earthquakes. As shown in Fig. 22(b), *R* of the TTLS increase as $Q(T_{RE|IM})$ increases. When $Q(T_{RE|IM})$ increases from 0.8 to 1.2, *R* of the TTLS under MSAS sequences at a *PGA* of 1.0 g increases by 34.6%.

It can be assumed that the post-recovery functionality follows a triangular distribution [62]. In this study, the upper limit, mode, and lower limit values are selected as 1.2, 1.0, and 0.8, respectively. 100 groups of uncertainty samples of the post-recovery functionality are generated using Monte-Carlo simulation. The effect of the uncertainty in post-recovery functionality on the seismic resilience of the TTLS is investigated, as shown in Fig. 23. It can be seen that those scatter points in the recovery processes exhibit a wider distribution as the restoration work progresses. Similarly, with the increase of *PGA*, the scatter points of *R* become more dispersed. For instance, when *PGA* is 1.0 g, the upper limit, 50%-percentiles, and lower limit values of *R* for the TTLS under the MSAS sequences increase by 29.1%, 10.6%, and 36.4%, respectively, when compared to those values observed under mainshocks. Therefore, these findings strongly indicate that the uncertainties associated with the post-recovery functionality should be considered in the seismic resilience assessment of TTLSs.

6. Conclusions

This paper develops a probabilistic framework for seismic resilience assessment that incorporates the consideration of aftershock effects. The seismic resilience of a TTLS under MSAS sequences is investigated using this framework. Several decision variables (functionality loss, recovery time, recovery process, and resilience indicator) are selected to quantitatively measure the post-event recovery capability. Notably, the influence of the uncertainties associated with these decision variables on the seismic performance of the TTLS is discussed. Based on the results, the following conclusions can be drawn:

(1) The probabilistic resilience assessment framework proposed in this paper emphasizes the effect of sequential earthquakes on the

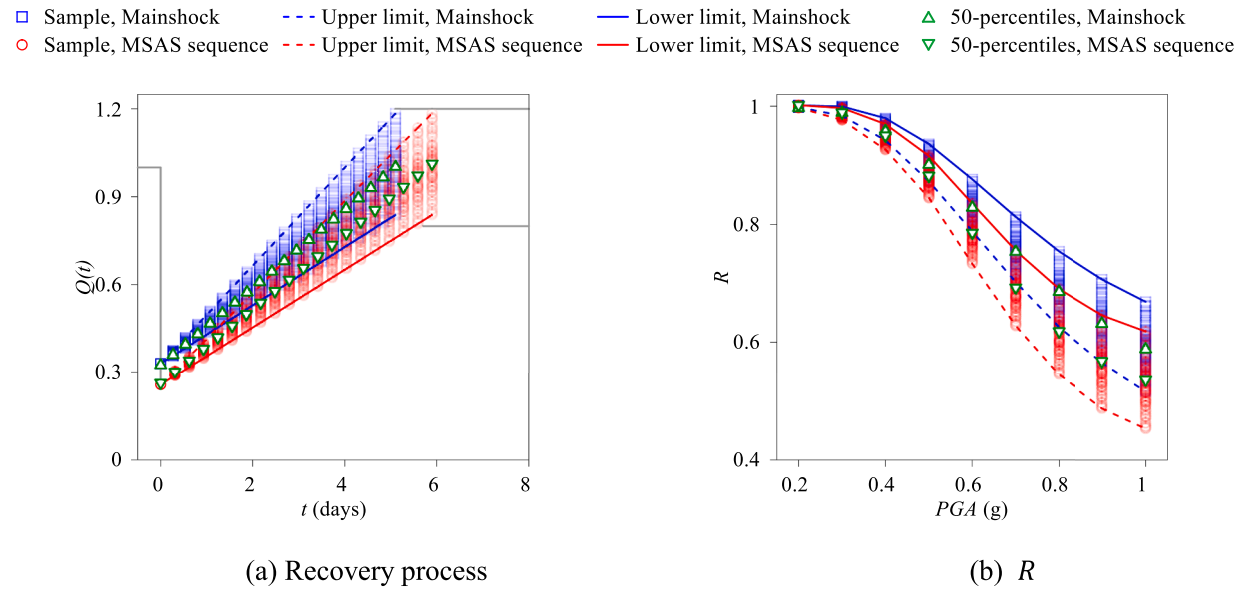


Fig. 23. Influence of the uncertainty in post-recovery functionality.

seismic resilience, specifically highlighting the incomplete functional recovery of structures during the time interval between mainshocks and subsequent aftershocks. The applicability of the framework in the seismic resilience assessment of TTLSs under sequential earthquakes is given importance.

(2) When the TTLS experiences minor or moderate damage after mainshocks, subsequent aftershocks result in significant additional damage to the TTLS, causing greater functionality losses and longer recovery times. However, once *PGA* exceeds 0.8 g, the TTLS becomes susceptible to major damage or collapse under the excitation of mainshocks, and the role of subsequent aftershocks becomes comparatively less influential.

(3) As *PGA* increases, both $L(E|IM)$ and $T_{RE|IM}$ increase, while R gradually decreases. Although the linear and trigonometric models yield similar predictions for R and ρ_Q, the latter can obtain a relatively lower χ_Q, indicating that a larger percentage of the recovery process can be completed over a shorter period of time around ρ_Q. Moreover, R gradually increase with the increase of $t_{gap}/T_{RE|IM,MS}$, indicating that as the functionality of the mainshock-damaged TTLS is restored, the recovery of the functionality of the TTLS after the subsequent aftershocks occurs at a faster pace.

(4) The uncertainties associated with the functionality loss, recovery time, and post-recovery functionality have a significant influence on the recovery process and resilience indicator of the TTLS. Their influence becomes more significant with the increase of earthquake intensity. Therefore, in the seismic resilience assessment of TTLSs, the uncertainties associated with these decision variables should be considered comprehensively.

This paper aims to present a resilience assessment framework that considers disaster chain, such as mainshock-aftershock sequences, and investigates its application in the post-disaster resilience assessment of engineering structures, specifically TTLS. For TTLS, it is important to note that it is not standalone and the investigation in the paper should be integrated in the seismic analysis of a power network. In future research, we will expand from a component level (transmission tower-line system) to a system level (power network), so as to comprehensively discuss the seismic damage and assess the post-disaster resilience of power networks under sequential earthquakes.

CRediT authorship contribution statement

Juncai Liu: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Li Tian:** Validation, Supervision, Funding acquisition, Formal analysis. **Meng Yang:** Writing – original draft, Investigation, Formal analysis. **Xiangrui Meng:** Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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